



Artificial Intelligence Fundamentals

(OPTIONAL) PROJECT ROADMAP

Why projects

Mix things up

Work on something that interests **you**

Get feedback along the way

Communicate your work

Grading: two different ones

Modality 1

Final Grade = oral exam on the course topics (material shared on Moodle)

- mostly, slides

Modality 2

Let X be the grade of the oral exam

If $X \geq 18$; then final grade = 30% project + 70% X ; else fail

In case of failure you can keep the same project for the next exam session

Non-attending students → Modality 1

Teams

Team composed by 1,2 or 3 members

- Try not to choose your close friends

Larger groups: for very challenging projects.

You are all expected to contribute

The project complexity should scale with the number of team members.

Find a **team contact reference**

- For communication with the teacher, submitting the report, etc.

Build!

Open project topic

- I can suggest some... but the choice is yours!

Built on GitHub:

- All members should be registered and part of the project
- All members should participate equally
- Self-contained project (I should be able to download and run it easily)

The project **MUST** contain methodologies studied during the course → the more, the better.

Report

Attention to details is key, length is not (make it 5-7 pages max),

The report should contain

- Introduction, related works, methodologies, results, conclusion
- Optional Appendix: team contributions, full experimental setup, additional results...
- Appendix and references do not count for the page limit

Latex template from NeurIPS 2025:

<https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip>

Present your project

You'll get the chance to present your project in December.

Main idea, dev. plan, issues, partial results, etc.

5-20 minutes team presentation or poster session depending on the teams number

All members should contribute to the preparation of the slides and oral presentation

The exact schedule will be decided with the team ref contacts

Basic tips to present

- Face your audience. I am not the only one in the room
- Don't put hands in pockets, in front of your mouth
- Do not rush it, be calm
- Speak clearly and "loud enough". Microphone can be provided. Use it if you need it!

Project submission

Team ref. will write to me: andrea.cossu@unipi.it

Always include [AIFProject] in the subject.

1. *Project proposal*: send an informal description of the activity and who-does-what. Subject to approval.
 - **Within October 16**
2. *Final material*: send the final project report (PDF) and the project presentation (PDF, pptx). Links to everything else (e.g., GitHub repo) need to be in the report.
 - **Within 7 days before the presentation day**

Project will remain valid forever.

Projects can easily be the starting pack for a master thesis, if interested

Project grading

I will evaluate the quality of the oral/poster presentation and the report.

- The report should contain all technical details
- The presentation should convey the project idea, the most important implementation points and the key empirical results

Project grades will be sent to team members within a week from the final presentation.

Given X the grade of the entire project, team members can receive less than X if they did not contribute equally / cannot properly explain what was done.

Before the official project delivery, you can revert back to exam modality 1, if you want.

There's no going back after the official delivery.

- What if I am not ok with my project grade?
- That should be a very rare case, if you do not cheat.
- Anyway, you might be asked to extend the project on your own for the next exam session.

Project assistance

A couple of lectures dedicated to project development

I will do my best to support you async

- I will not be able to provide one-to-one help to everyone.

The evaluation will take that into consideration

That sounds convoluted,
why should I do it?

Open to suggestions

In the final notes of the mandatory survey (before the final exam) you can write your thoughts on how to better organize the project/what was your experience

I am very open to suggestions during the course as well

Suggestions will be taken into consideration for the **next** year

Project ideas: go to the gym

<https://gymnasium.farama.org/>

The most popular set of environments, ranging from very simple to quite challenging.

Many unofficial gym environment you can use on the web.

Objective:

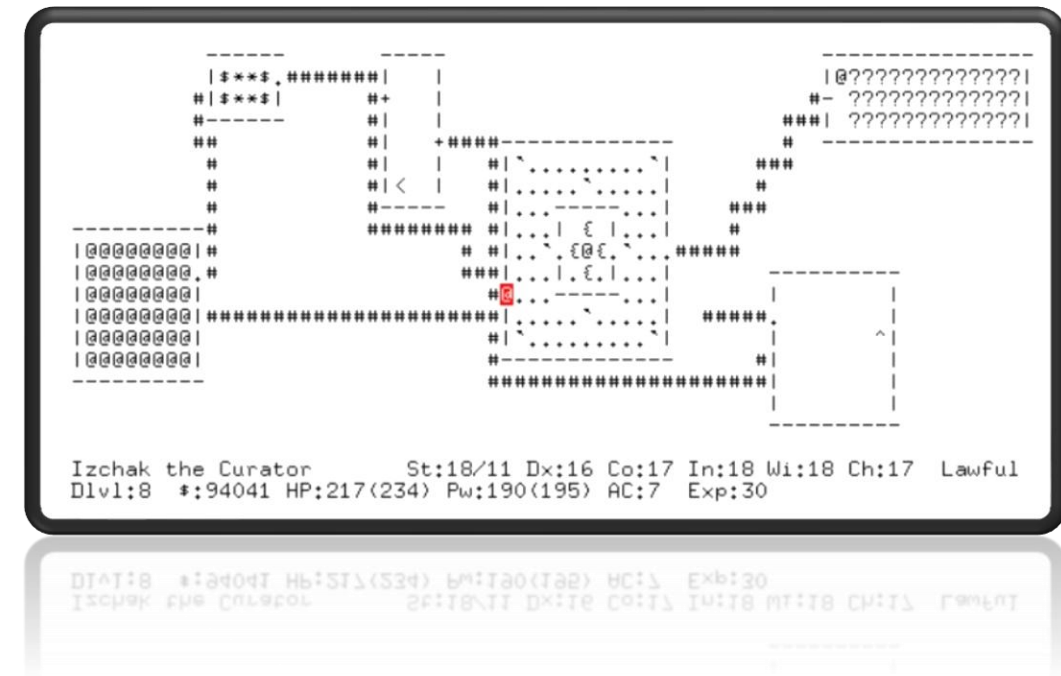
- try out a few algorithms (search, evolutionary, planning, etc...) to solve the task.
- Try out a few environments

We will see some examples during our in-class labs

Project ideas: NetHack

Useful references:

- [YouTube video](#) (introductory video)
- [The NetHack Learning Environment](#) (paper)
 - [NetHack-LE/nle](#) (github)
- [NetHack Challenge](#) (with python notebooks)
- [Alt.org server](#) (play NetHack)



<https://rockt.ai/>

Tim Rocktäschel – Open-Endedness team leader @ Deep Mind

Project ideas: NetHack

Smart pathfinding: implement an agent able to reach one or multiple goals while minimizing the number of steps required, being aware of its surroundings (e.g. the actions of monsters). The agent should be able to react to dangerous situations while performing its plan;

Advanced combat system: define a module to manage the combat system. The agent should be able to correctly use the different types of weapon available (e.g. swords, bows, wands).

Inventory management: define an agent able to manage its inventory.

Floor recognition: NetHack provides different types of floors and rooms (e.g. shops). The task is to implement a module that can recognize the current floor/room, in order to behave accordingly;

State representation: Represent the states of the game, possibly taking into consideration also n previous states. It can be done using several methodologies, e.g. Markov Chains, Bayesian Networks, etc. This encoding can be used to decide the action to take, or for next-state prediction;

Logic Programming: Define a knowledge base using Prolog, to represent the game dynamics, by exploiting the information present on the NetHack Wiki; it can be done in a custom environment, defined with MiniHack.

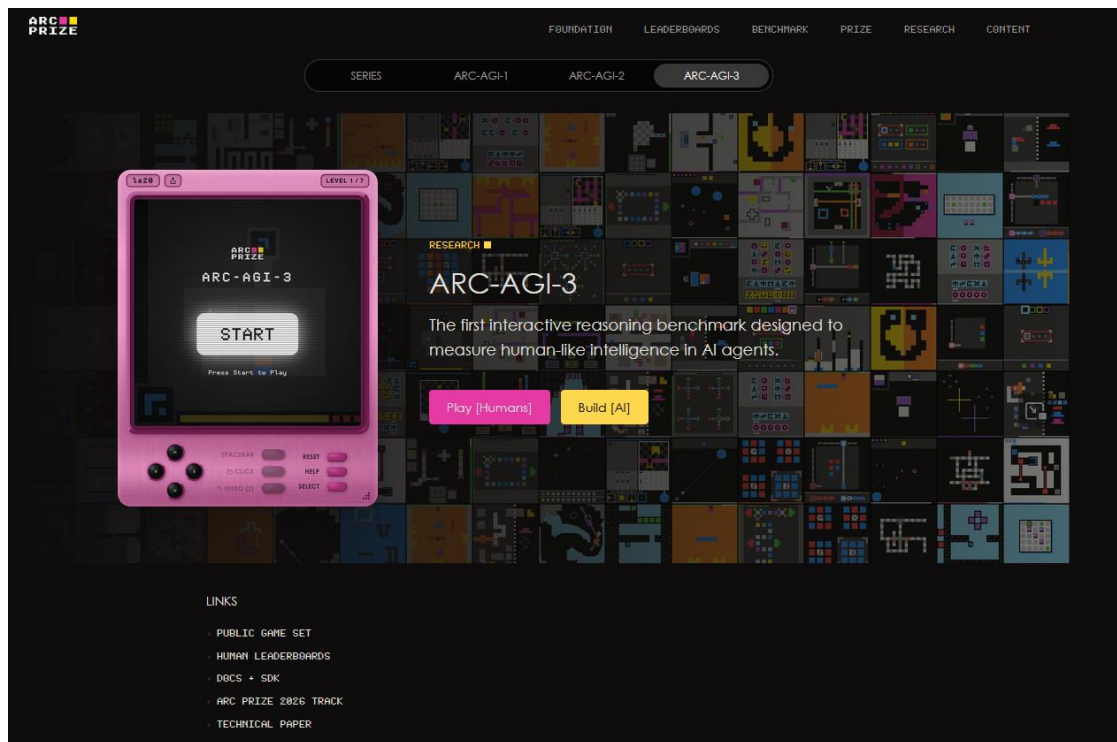
Project ideas: Artificial Life

You can create digital worlds, and explore them via digital evolution / search methods

- Manual search is also allowed 😊

[BioMaker](#)

[Particle Lenia](#)



ARC-AGI

<https://arcprize.org/arc-agi/3>

We introduce ARC-AGI-3, an interactive benchmark for studying agentic intelligence through novel, abstract, turn-based environments in which agents must explore, infer goals, build internal models of environment dynamics, and plan effective action sequences without explicit instructions. Like its predecessors ARC-AGI-1 and 2, ARC-AGI-3 focuses entirely on evaluating fluid adaptive efficiency on novel tasks, while avoiding language and external knowledge. ARC-AGI-3 environments only leverage Core Knowledge priors and are difficulty-calibrated via extensive testing with human test-takers. Our testing shows humans can solve 100% of the environments, in contrast to frontier AI systems which, as of March 2026, score below 1%. In this paper, we present the benchmark design, its efficiency-based scoring framework grounded in human action baselines, and the methodology used to construct, validate, and calibrate the environments.

Project ideas: the hands game

Each player starts with one finger per hand.

Players play in turn, one action per turn

Action: choose one of your hands, attack one of the opponent's hand.

Result: the opponent hand will have as many fingers as the sum of both hands, modulo five

If one hand reaches exactly 5 fingers, it dies. When both hands die, the other player wins.

If one player is left with one hand where there are exactly 2 or 4 fingers, the player can split that hand in two hands with half the fingers in each hand

Extend to >2 players?

Project ideas: hybrid systems

Can you leverage GenAI as a CoPilot for classical AI algorithms?

- E.g., as a heuristics-generator (you'll understand later on)

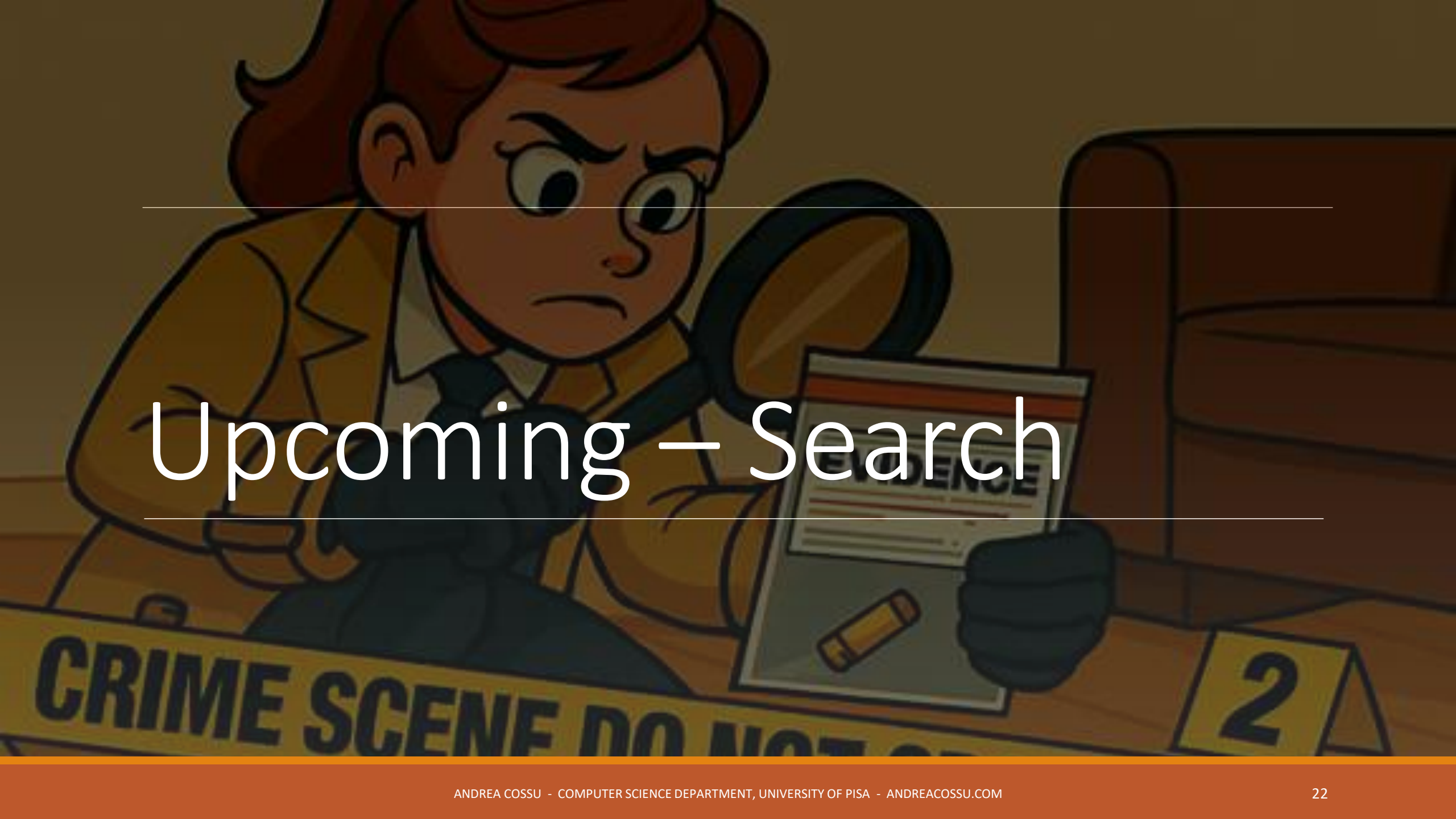
Can GenAI simulate classical AI algorithms?

More in general, how can you use GenAI to empower classical AI algorithm?

A comprehensive evaluation of different techniques and/or development of new ideas are welcome.

Project ideas: Symbolic vs. Generative AI



A cartoon illustration of a detective with brown hair and a determined expression, wearing a tan trench coat and blue gloves. The detective is holding a magnifying glass over a piece of evidence labeled 'EVIDENCE' which contains a yellow pencil. In the foreground, a yellow evidence marker with the number '2' is visible. A yellow banner with the text 'CRIME SCENE DO NOT CROSS' is partially visible at the bottom. The background is a dark, textured brown.

Upcoming – Search